ELEG 5633: Detection and Estimation
Homework 5

1. We observe the i.i.d samples X_0, X_1, ..., X_{N-1} from the distribution $\mathcal{N}(\mu, \sigma^2)$. Consider the binary hypethesis testing

$$\begin{aligned} H_0 : &\quad \mu = \mu_1 \\ H_1 : &\quad \mu = \mu_2 \end{aligned}$$

   (a) Design the NP test under the constraint $P_{\text{FA}} < \alpha$.

   (b) Derive P_{D}, which should be expressed as a function of P_{FA}.

   (c) Set $\sigma^2 = 1$, $\mu_1 = 1$, and $\mu_2 = 2$. If we want $P_{\text{D}} \geq 0.9$ and $P_{\text{FA}} \leq 0.01$, how many samples do we need in the detection?

   (d) Set $\sigma^2 = 1$, $\mu_1 = 1$, and $\mu_2 = 2$. Use Matlab to plot in the same figure the ROC curves when $N = 1$, $N = 10$, and $N = 20$.

2. We observe the i.i.d samples X_0, X_1, ..., X_{N-1} from the distribution $\mathcal{N}(0, \sigma^2)$. Consider the binary hypethesis testing

$$\begin{aligned} H_0 : &\quad \sigma^2 = \sigma_0^2 \\ H_1 : &\quad \sigma^2 = \sigma_1^2 \end{aligned}$$

   where $\sigma_1^2 > \sigma_0^2$.

   (a) Design the NP test under the constraint $P_{\text{FA}} = 0.001$.

   (b) Derive P_{D}

3. Consider an exponential distribution with pdf

$$p_i(x) = \frac{1}{\mu_i} \exp\left(-\frac{x}{\mu_i}\right), \quad x \geq 0$$

   for $i = 0, 1$. Find the KL divergence $D(p_1 \| p_0)$, and caculate $D(p_1 \| p_0)$ when $\mu_0 = 1$ and $\mu_1 = 2$.